

altairsim — 88-UIO: two boards in one

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Altair 8K BASIC 3.2 loads from a cassette through one **88-UIO** board.

```
cd examples/uio
altairsim uio.toml
```

The machine stops at the BASIC prompts. Press Enter at **MEMORY SIZE?** and at **TERMINAL WIDTH?**. Type **Y** at **WANT SIN-COS-TAN-ATN?**. BASIC then prints:

```
10312 BYTES FREE

ALTAIR BASIC VERSION 3.2
[EIGHT-K VERSION]

OK
```

What is different

The built-in **basic8k** machine needs two boards. An 88-2SIO is the console. An 88-ACR is the cassette. This machine has one board, the MITS 88-UIO, which does both jobs.

- The 6850 serial port of the 88-UIO responds to port 0x10. This is where Port A of an 88-2SIO responds. The serial port is the console.
- The cassette section responds to port 0x06. This is where an 88-ACR responds. The cassette section is the tape.

Both sections use the standard ports. Therefore the MITS bootstrap **LDR8K32.HEX** runs without a change. 8K BASIC does not know that one board does both jobs. The sense switches are **0x8C**. The header of the loader gives this value: A15 selects load from cassette, and A11 and A10 select an 88-2SIO terminal.

Try it

Type a program at the `OK` prompt:

```
10 PRINT 6*7
20 PRINT "TAPE OK"
RUN
```

An 88-UIO can do things that an 88-ACR cannot. Press `Ctrl-E` to get the monitor, then type:

```
SHOW uio0          ; show both sections: serial_port, port, standard, motor
SET uio0 standard=kansas ; use the Kansas City standard (2400/1200) for the modem
REWIND uio0:tape    ; put the tape back at the start
```

`Ctrl-E` returns you to the `altairsim>` monitor at any time.

Files

File	What it is
<code>uio.toml</code>	The machine file: one 88-UIO, 16K of RAM, the front panel and the console.
<code>8K BASIC Ver 3-2.tap</code>	Altair 8K BASIC 3.2, a cassette image from the period.
<code>LDR8K32.HEX</code>	The 8K bootstrap loader. The machine file puts it in memory at address 0.

The processor runs at full speed, and the tape loads in about one second. To run at the 2 MHz speed of the real machine, type `SET cpu0 clock_hz=2000000`. A 300-baud cassette then takes about 90 seconds to load.